OPIHI master dataset: verification and corrections

Independent check of the Day 1 cleaning against the raw datasheets · GroupC_AI_Data_Sprint · 9 November 2026

Summary

Every count in the AI-built master was traced back to its raw spreadsheet cell, and every flag, exclusion and name mapping was tested independently. **No counts were lost or misread.** Five corrections were applied, all to names, categories or record structure; the total of all counts is unchanged. The corrected tables are *opihi_master_completed*, *opihi_surveys_completed* and *opihi_samples_completed*, with every edit listed in *opihi_change_log*.

Item	Original master	After corrections
Detections (rows)	22,588	22,586 (2 duplicate rows merged)
Total of all counts	-	Unchanged
Surveys	252	252
Samples	7,216	7,216
Site names in use	43	42 (Sandy Beach West merged into Sandy Beach)
Records traced to raw cells	-	22,588 of 22,588 (families A, B and C)

Checks performed

Each check was run in R against the raw Excel cells or recomputed from the processed tables, not taken from the AI's own verification table.

Area	What was tested	Result
Families A and B (legacy and modern sheets)	Each master row matched to its raw cell by file, row and column	22,241 of 22,241 match
Family C (classroom group files)	Forward (master to raw) and reverse (every non-zero raw count present in master)	347 of 347 in both directions. The 2025 P1 tab and the Waipuilani file keep their labels in column B, a layout difference only
Master vs sample table	Count totals per sample	Agree in all 7,216 samples; the n_taxa difference was investigated and explained
Updated Waipuilani file	Team's newer copy compared cell by cell with the copy the checks used	Cosmetic only: line-break characters and recalculated proportion formulas. No count or name changed
QC flags (296)	Every flag points to a real record; sample-table flag column vs flag file	All 239 sample-level rows and 57 survey-level rows resolve. Flag counts agree for all 5 sample flag types
points_over_25 rule	Rule recomputed from point sums	230 samples over the limit; all flagged, none flagged wrongly. Split: 174 legacy (Family A) + 56 modern (Family B) quadrats, matching the handoff guide's 174
placeholder_time_removed (45)	Original values and current state	All 45 were the 00:00-01:00 template default; all now blank
site_corrected_by_coordinates (3)	Each block opened in the raw file	Master site is correct for all three (S0143 Onekahakaha, S0167 Leahi, S0210 Kamaole Beach Park)

Area	What was tested	Result
Site label of every legacy survey	Master site vs the location written inside the survey's own block	227 of 227 match; 0 differ
Compiled-file cross-check	234 trips said to be in master; 18 said to have no species data	All 234 found by site and date. All 18 rows confirmed empty in the compiled workbook
Exclusions log (427)	Every entry accounted for by rule and source	309 blank template rows, 109 no taxon data, 6 invalid taxon names, 1 example row, 1 summary row, 1 survey without metadata
Taxon names	Misspellings in S0251; one taxon per sample	All misspellings mapped to the right species. 2 duplicate taxa found and merged (below)
Site names	Spelling variants; any name on two islands	No name on two islands. One pair found to be the same site

Corrections applied

These are recorded row by row in *opihi_change_log*. None changes the total of all counts.

Change	Rows	Reason
S0251: site (Leahi) and transect recovered	Survey S0251	Recoverable from the raw classroom file; left blank by the cleaning
Dendropoma gregaria renamed *Dendropoma gregarium*	215	The AI used the majority raw spelling; the accepted spelling was chosen. All three raw spellings already pointed to one taxon
Category of 'Unknown fish': Invertebrate to Vertebrate	1	Fish are vertebrates; the category list had no vertebrate option, so a Vertebrate category was added
Duplicate taxon rows merged in S0251-003: *Sargassum aquifolium* 1 + 2 = 3, *Turbinaria ornata* 1 + 1 = 2	2 rows removed	A misspelling and the correct name were entered on separate sheet lines (confirmed in the raw file). Counts summed; both raw spellings and both source cells kept. P1 sheet totals (4 + 22; 2025: 6) match master
Site 'Sandy Beach West' renamed 'Sandy Beach'	54 (surveys S0101, S0214)	Same site under two names, confirmed by the reviewer

Findings that needed no change to the data

- **Legacy parser reads the neighbouring block's location.** In all three site_corrected_by_coordinates flags, the 'labelled' site came from the block next door. The coordinate rule then restored the correct site, so master was right; the full block check (227 of 227) shows the bug never misplaced a survey. It affects flag wording only.
- **Legacy coordinates were copied from a site list, not measured.** Each flagged survey sat exactly (0.00 km) on a reference point, so coordinates are not independent evidence of location.
- **A flag contradicts its own detail.** The no_metadata_match flag (2024 Mauliola, 2024-03-26) has the same 'matched within 7 days' detail as metadata_date_mismatch.
- **unmatched_names is empty by construction.** The crosswalk was built from the same data it was applied to (300 spellings to 215 names), so it cannot flag problems in existing data; it will only catch new spellings.
- **Two taxa written in one cell** (Waipuilani 2026): the *Sargassum aquifolium* + *S. polyphyllum* row was filed as *Sargassum* spp.; the *Pterocladiella* row has no counts.
- **Smaller survey flags look reasonable:** 5 time_assumed_pm (unlabelled afternoon times consistent with low tide) and 2 no_site_or_date_in_file (the 2023 and 2025 classroom files).

- **Teacher ID T043** appears on five islands, so it is almost certainly a shared or placeholder ID rather than one person.

Decision for each QC flag type

Every flag type was checked against the data or the raw file. No flag required a change to the counts; the only fixes (placeholder times, S0251) were already applied.

Flag	n	Evidence	Decision
points_over_25	230	Rule recomputed exactly; 174 legacy + 56 modern quadrats	**Keep, flagged.** Fine for detection; exclude or note when computing percent cover
total_points_mismatch	10	All in the 2024 datasheet. In 5 the cells sum to exactly 25 while the typed total is wrong (13-26); in 4 the cells exceed 25 (also flagged over 25); in 1 the cells sum to 18 against a typed 25	**Keep the cell values**; the typed totals are unreliable
placeholder_time_removed	45	All were the 00:00-01:00 template default; all now blank	**Fix (applied)**
site_corrected_by_coordinates	3	Raw blocks checked; 227 of 227 legacy sites match their own block	**Keep** master's site; flag text names the wrong original label
time_assumed_pm	5	Afternoon reading fits the low tide times	**Keep**
no_site_or_date_in_file	2	S0251 site and transects recovered; nothing to recover for S0252	**Fix S0251 (done); keep S0252** but exclude it from site and time analyses
metadata_date_mismatch	1	Matched on file and site within 7 days; data-row date kept	**Keep**
no_metadata_match	1	Detail says it did match within 7 days, same as above	**Keep**; flag name is wrong
date_inferred	1	S0246-037 sits between transect 3 and 4 neighbours on the same date	**Keep**
header_inferred	1	S0128-011 inferred as transect 2, quadrat 1, between T1 Q10 and T2 Q2	**Keep**
group_label_mismatch	1	In S0252, which has no site	**Keep**

Limitations kept in the final data

- **2020 block08 stays excluded.** It holds real data (4 taxa, 20 quadrats, 38 detections) but no site and no reliable date; the file order only bounds it to January 2020 or earlier.
- **S0252 has no site** (2025 classroom file) and cannot be mapped.
- **S0251 and S0252 are not split by tab (P1/P2).** S0251's charts name Diamond Head on both tabs and neither tab has a date, so there is no evidence of two separate visits (P most likely means class period). S0252 has no site or date on either tab. Both keep the year only.
- **The *Turbinaria* merge (1 + 1)** assumes two separate records; a single observation entered twice cannot be ruled out without the field notes.

Notes for the team

- **Re-running the team pipeline** rebuilds the original files only. The corrections live in the _completed files and are re-applied by the verification script, so rerun it after any pipeline rerun.
- **Repository:** raw datasheets were removed from GitHub and added to .gitignore; they remain in the commit history.